

AS2050* — Undergraduate Aerodynamics

Lecture Plan

3rd Semester (July–November 2026)

Instructor:

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B Slot • Mon 09:00, Tue 08:00, Fri 11:00 (Lectures) • Wed 13:00 (Tutorial)

Last Updated: July 25, 2026

Course Information

Course Code: AS2050*
Course Title: Undergraduate Aerodynamics
Department: Aerospace Engineering
Institution: Indian Institute of Technology Madras
Semester: 3rd Semester (July–November 2026)
Slot: B

Key Dates (B Slot — from the Institute Academic Calendar)

Event	Date
Commencement of classes	Monday, 27 July 2026
Quiz–I (B/M)	Tuesday, 1 September 2026
Quiz–II (B/M)	Tuesday, 6 October 2026
Last Instructional Day	Friday, 6 November 2026
End–Semester Exam (B)	Friday, 13 November 2026

Structure at a Glance

- **40 lectures** (Mon / Tue / Fri) and **14 tutorials** (Wed), net of holidays and quiz days.
- Three teaching windows separated by the two quizzes, each closing with a dedicated buffer/review lecture.
- Wednesday is reserved for tutorials (problem-solving, TA-supported).
- Holidays already absorbed: Milad (Wed 26 Aug), Ganesh Chaturthi (Mon 14 Sep), Gandhi Jayanti (Fri 2 Oct), Dussehra (Tue 20 Oct).

Course Spine

Block	Theme	Lectures
B0	Foundations & review (math, kinematics, governing laws)	L1–L3
B1	Potential-flow toolkit (elementary flows, complex potential)	L4–L8
B2	Vorticity & circulation (Helmholtz, Kelvin, Biot–Savart)	L9–L12
B3	Two-dimensional airfoils (Kutta condition, Joukowski)	L13–L15
	<i>Quiz–I — covers L1–L15</i>	1 Sep
B4	Thin-airfoil theory, panel methods, experimental data	L17–L23
B5	Propellers (momentum & blade-element theory)	L24–L26
	<i>Quiz–II — covers L17–L27</i>	6 Oct
B6	Finite wings (lifting-line theory, induced drag, swept/delta)	L29–L35
B7	Whole aircraft & synthesis	L36–L40

Window 1 — Up to Quiz–I (Tue 1 Sep 2026)

Date	Session	Topic
Mon 27 Jul	L1	Introduction to aerodynamics; lift/drag/moment; applications & history; course policies, tutorial/TA structure, phone policy
Tue 28 Jul	L2	<i>Review</i> — vector calculus: grad/div/curl; divergence & Stokes' theorems
Wed 29 Jul	T1	Tutorial: vector-calculus problems
Fri 31 Jul	L3	<i>Review</i> — fluid kinematics: continuum, material derivative, Eulerian/Lagrangian, streamlines
Mon 03 Aug	L4	Conservation laws (continuity, momentum); incompressibility; irrotationality; velocity potential
Tue 04 Aug	L5	Stream function; Laplace equation; boundary conditions; superposition principle
Wed 05 Aug	T2	Tutorial: governing-equations problems
Fri 07 Aug	L6	Elementary flows I — uniform flow, source/sink, doublet
Mon 10 Aug	L7	Elementary flows II — Rankine half-body; non-lifting flow over a cylinder
Tue 11 Aug	L8	Complex potential & conformal mapping (analytic tool, after elementary flows)
Wed 12 Aug	T3	Tutorial: elementary-flows & superposition
Fri 14 Aug	L9	Vorticity & circulation; the point vortex
Mon 17 Aug	L10	Lifting flow over a cylinder; Kutta–Joukowski theorem $L = \rho V_\infty \Gamma$
Tue 18 Aug	L11	Vortex dynamics — Helmholtz's vortex theorems; Kelvin's circulation theorem
Wed 19 Aug	T4	Tutorial: vorticity, circulation, Kutta–Joukowski
Fri 21 Aug	L12	Biot–Savart law; vortex filament; vortex sheet
Mon 24 Aug	L13	Airfoil nomenclature & characteristics; c_l, c_d, c_m ; pressure distribution; aerodynamic centre
Tue 25 Aug	L14	Kutta condition; starting vortex; generation of circulation on an airfoil
Wed 26 Aug	—	<i>Holiday — Milad (no tutorial)</i>
Fri 28 Aug	L15	Joukowski transformation → airfoil lift; synthesis
Mon 31 Aug	L16	Quiz–I buffer — review & worked problems
Tue 01 Sep	QUIZ–I	Covers L1–L15

Window 2 — Quiz–I to Quiz–II (Tue 6 Oct 2026)

Date	Session	Topic
Wed 02 Sep	T5	Tutorial: Quiz–I solutions + airfoil problems
Fri 04 Sep	L17	Thin-airfoil theory I — vortex sheet on the camber line; fundamental equation
Mon 07 Sep	L18	Thin-airfoil theory II — symmetric airfoil: $c_l = 2\pi\alpha$; a.c. at $c/4$
Tue 08 Sep	L19	Thin-airfoil theory III — cambered airfoil: Fourier-series solution; c_l, c_m
Wed 09 Sep	T6	Tutorial: thin-airfoil problems I
Fri 11 Sep	L20	Thin-airfoil theory IV — moments; centre of pressure vs. aerodynamic centre; examples
Mon 14 Sep	—	<i>Holiday — Ganesh Chaturthi</i>
Tue 15 Sep	L21	Method of singularities; source/vortex panel methods — concept & discretisation
Wed 16 Sep	T7	Tutorial: thin-airfoil problems II
Fri 18 Sep	L22	Vortex panel method — implementation; introduction to computational aerodynamics
Mon 21 Sep	L23	Experimental airfoil characteristics — Reynolds-number effects, stall, drag polar
Tue 22 Sep	L24	Propellers I — momentum (actuator-disc) theory; thrust; ideal efficiency
Wed 23 Sep	T8	Tutorial: panel-method / airfoil-data problems
Fri 25 Sep	L25	Propellers II — blade-element theory (uses 2D airfoil data)
Mon 28 Sep	L26	Propellers III — combined blade-element–momentum (BEM); performance charts
Tue 29 Sep	L27	Consolidation — thin-airfoil + propeller worked problems
Wed 30 Sep	T9	Tutorial: propeller (BEM) problems
Fri 02 Oct	—	<i>Holiday — Gandhi Jayanti</i>
Mon 05 Oct	L28	Quiz–II buffer — review
Tue 06 Oct	QUIZ–II	Covers L17–L27

Window 3 — Quiz–II to End-Semester

Date	Session	Topic
Wed 07 Oct	T10	Tutorial: Quiz–II solutions + finite-wing introduction
Fri 09 Oct	L29	Finite wings — downwash; induced angle of attack; induced drag; wingtip vortices
Mon 12 Oct	L30	Prandtl's lifting-line theory I — the fundamental equation
Tue 13 Oct	L31	Lifting-line theory II — elliptical lift distribution; minimum induced drag; span efficiency
Wed 14 Oct	T11	Tutorial: lifting-line problems I

Date	Session	Topic
Fri 16 Oct	L32	Lifting-line theory III — general distribution (Fourier); aspect-ratio effects
Mon 19 Oct	L33	Effect of geometry — aspect ratio, taper, twist on lift & induced drag
Tue 20 Oct	—	<i>Holiday — Dussehra</i>
Wed 21 Oct	T12	Tutorial: lifting-line / aspect-ratio problems II
Fri 23 Oct	L34	Elements of lifting-surface theory; vortex-lattice concept
Mon 26 Oct	L35	Swept & delta wings — sweep effects; leading-edge vortices
Tue 27 Oct	L36	Flow past bodies in incompressible flow; drag breakdown (pressure vs. friction)
Wed 28 Oct	T13	Tutorial: swept/delta + whole-aircraft problems
Fri 30 Oct	L37	Lift, drag & moment of the whole airplane; aircraft drag polar; L/D
Mon 02 Nov	L38	Putting it together: airfoil → wing → aircraft; applications
Tue 03 Nov	L39	Course synthesis & outlook (compressibility preview); End-sem review I
Wed 04 Nov	T14	Tutorial: comprehensive end-semester problem set
Fri 06 Nov	L40	End-sem review II — worked problems & Q&A (<i>Last Instructional Day</i>)
Fri 13 Nov	END-SEM	Covers the entire course

Assessment

Component	Timing	Weight
Continuous Assessment	Weekly tutorial quizzes + assignments	20%
Software Exercise	Individual (aerodynamics software)	5%
Group Presentation	Group (15 min)	5%
Quiz–I	Tue 1 Sep 2026 (L1–L15)	15%
Quiz–II	Tue 6 Oct 2026 (L17–L27)	15%
End-Semester Exam	Fri 13 Nov 2026 (entire course)	40%
Total		100%

Text References

1. Anderson, J. D. *Fundamentals of Aerodynamics*, 7th ed. (*primary*)
2. Kuethé, A. M., & Chow, C.-Y. *Foundations of Aerodynamics*, 5th ed.
3. Kundu, P. K., Cohen, I. M., Dowling, D. R., & Capecelatro, J. *Fluid Mechanics*, 6th ed.
4. Acheson, D. J. *Elementary Fluid Dynamics*.

Notes

- Quiz days occupy that day's B-slot session; no new content is scheduled on a quiz day.
- A buffer/review lecture (L16, L28) precedes each quiz to absorb minor slippage without cascading.
- Tutorials trail the corresponding lecture block so that problem-solving follows the theory.
- If the plan runs ahead, buffer lectures convert to additional problem sessions rather than new content.